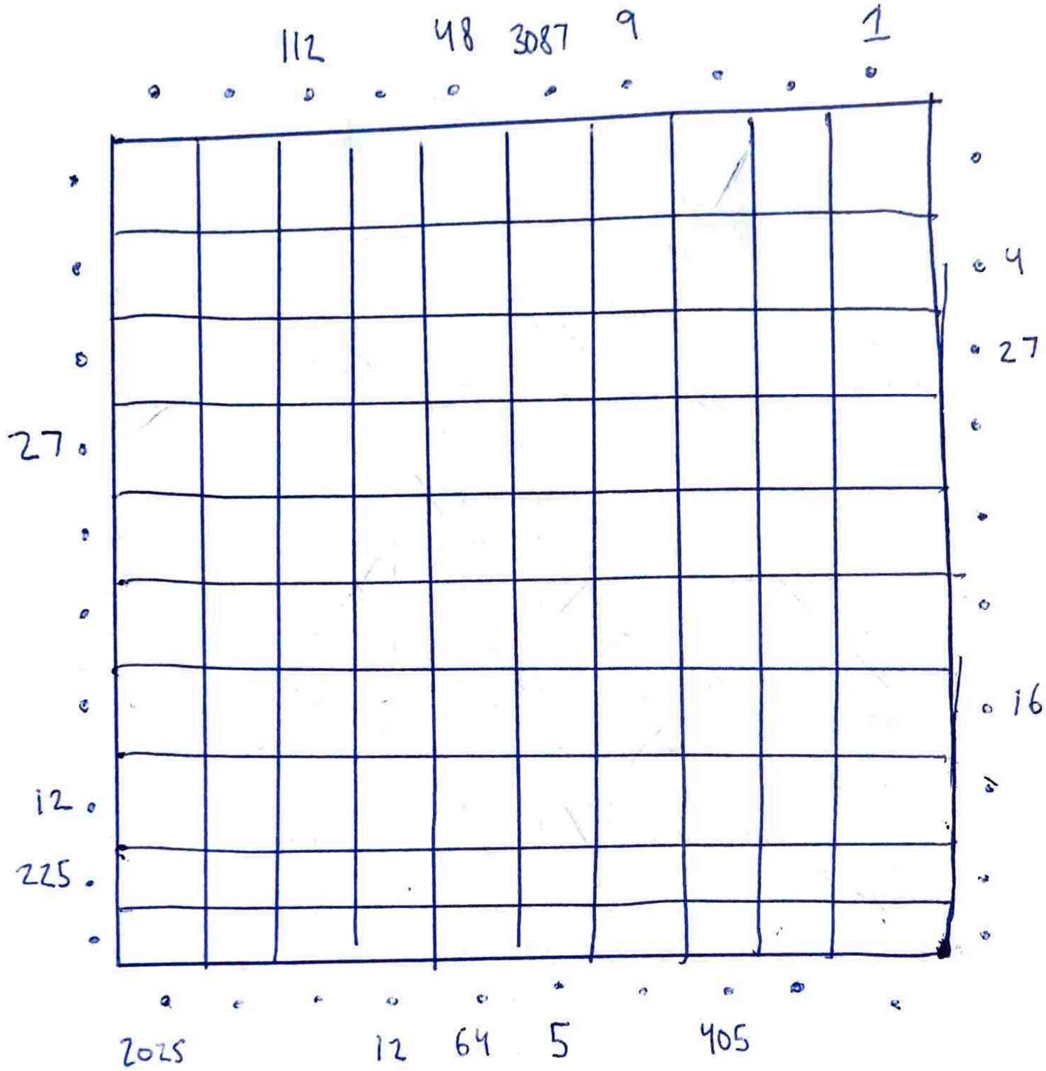
Jane Street Puzzle March 2025

Hall of Mirrors 3

Dylan Edwards



Q: The perimeter of a 10-by-10 square field is surrounded by lasers pointing into the field. Some of the lasers have numbers beside them. Place diagonal mirrors in some of the cells so that the product of the segment lengths of a laser's path matches the clue numbers. Mirrors may not be placed in orthogonally adjacent cells.

Once finished, determine the missing clue numbers for the perimeter, and calculate the sum of these clues for each side of the square. The answer to this puzzle is the product of these four sums.

Solution

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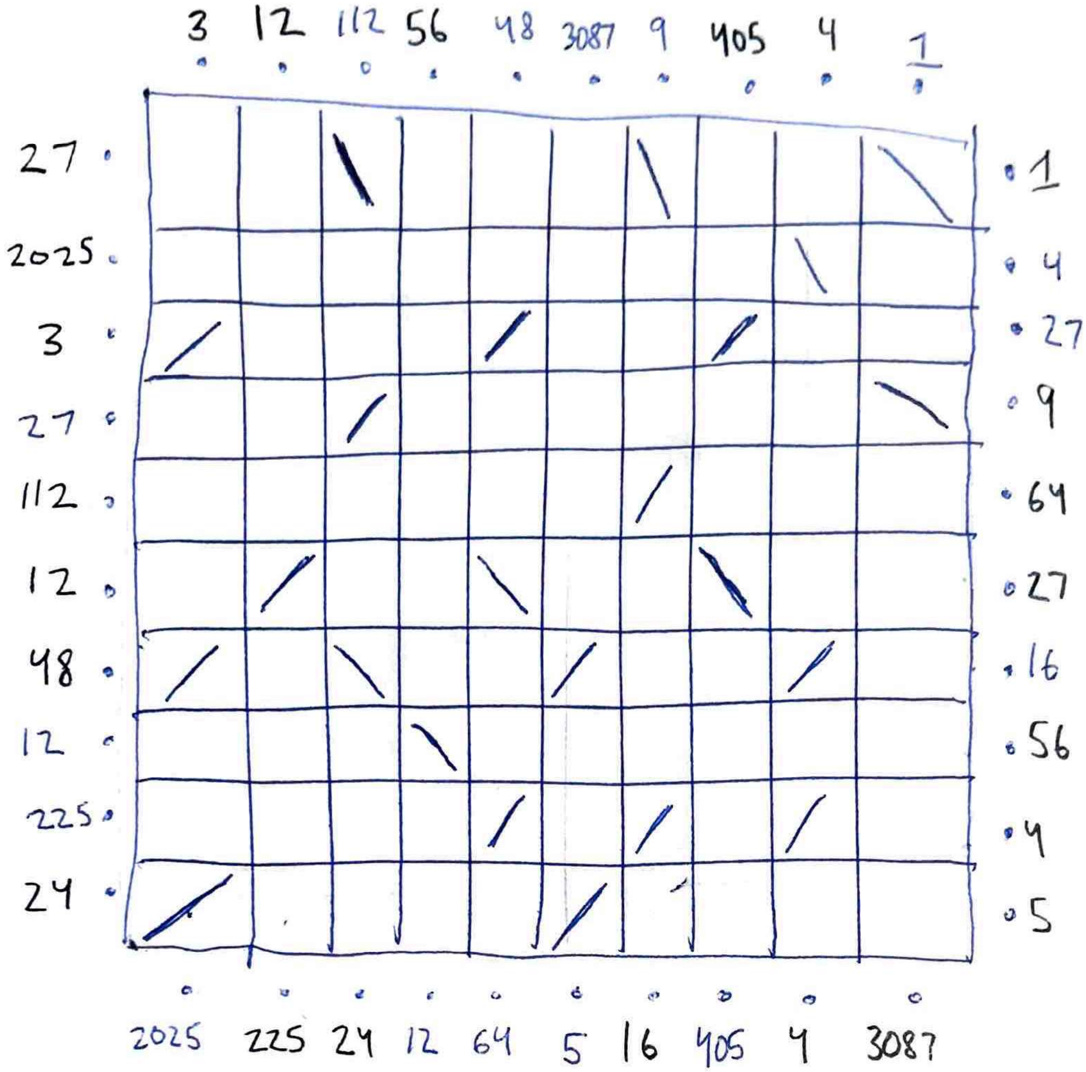

Missing Numbers

Left: $27 + 2025 + 3 + 112 + 12 + 48 + 24 = 2251$

Top: $3 + 12 + 56 + 405 + 4 = 480$

Right: $1 + 9 + 64 + 27 + 56 + 4 + 5 = 166$

Bottom: $225 + 24 + 16 + 4 + 3087 = 3356$

$2251 \times 480 \times 166 \times 3356 = \boxed{601931086080}$